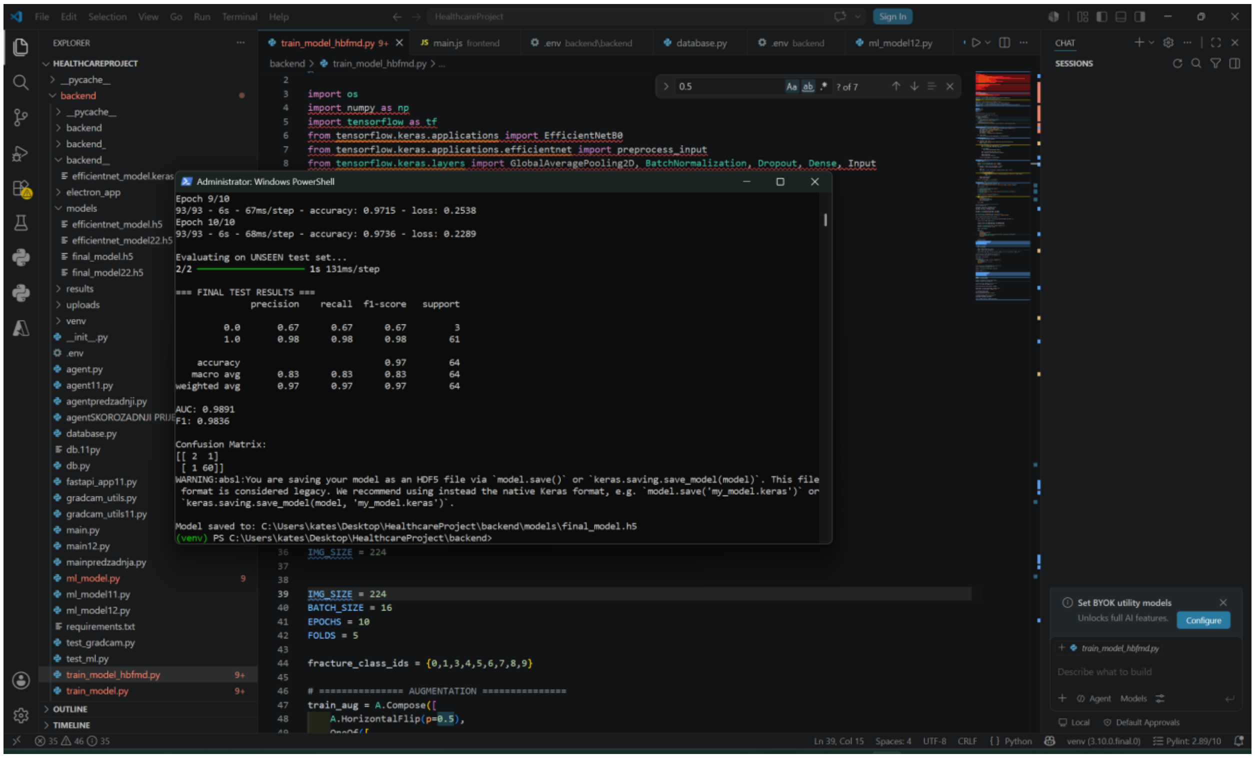
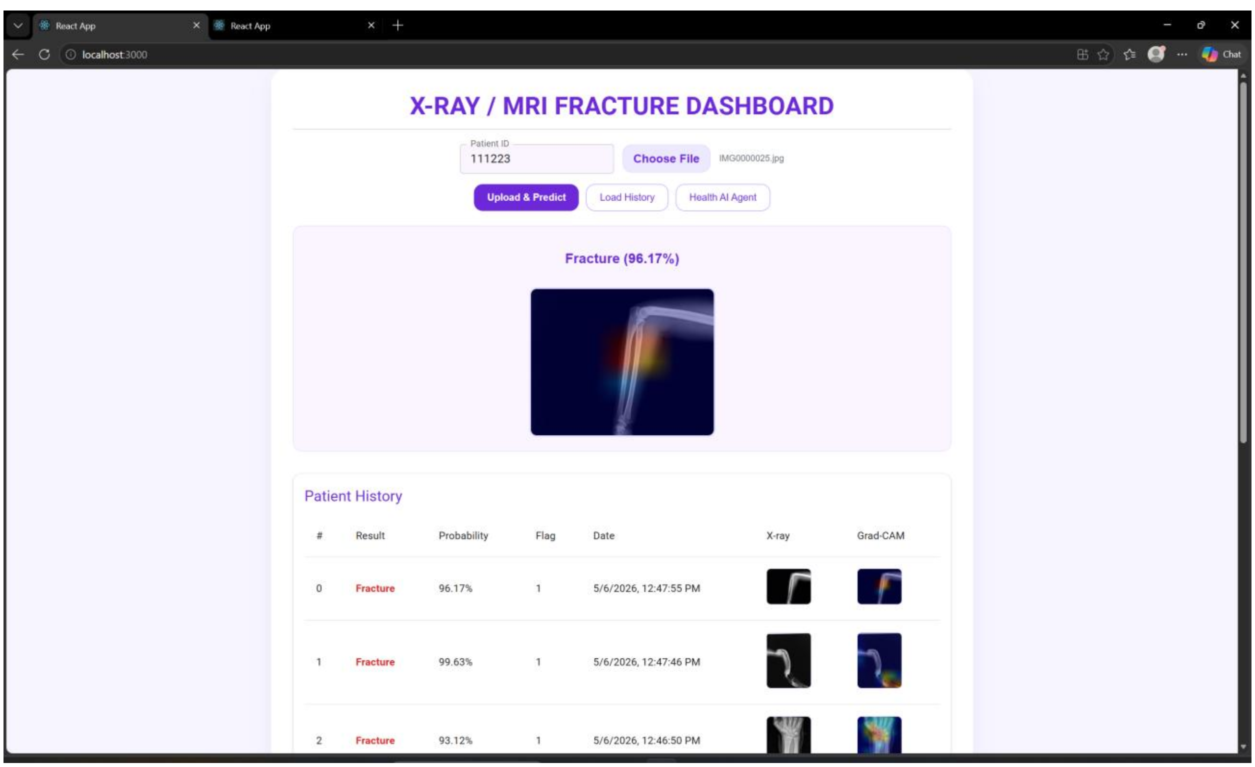
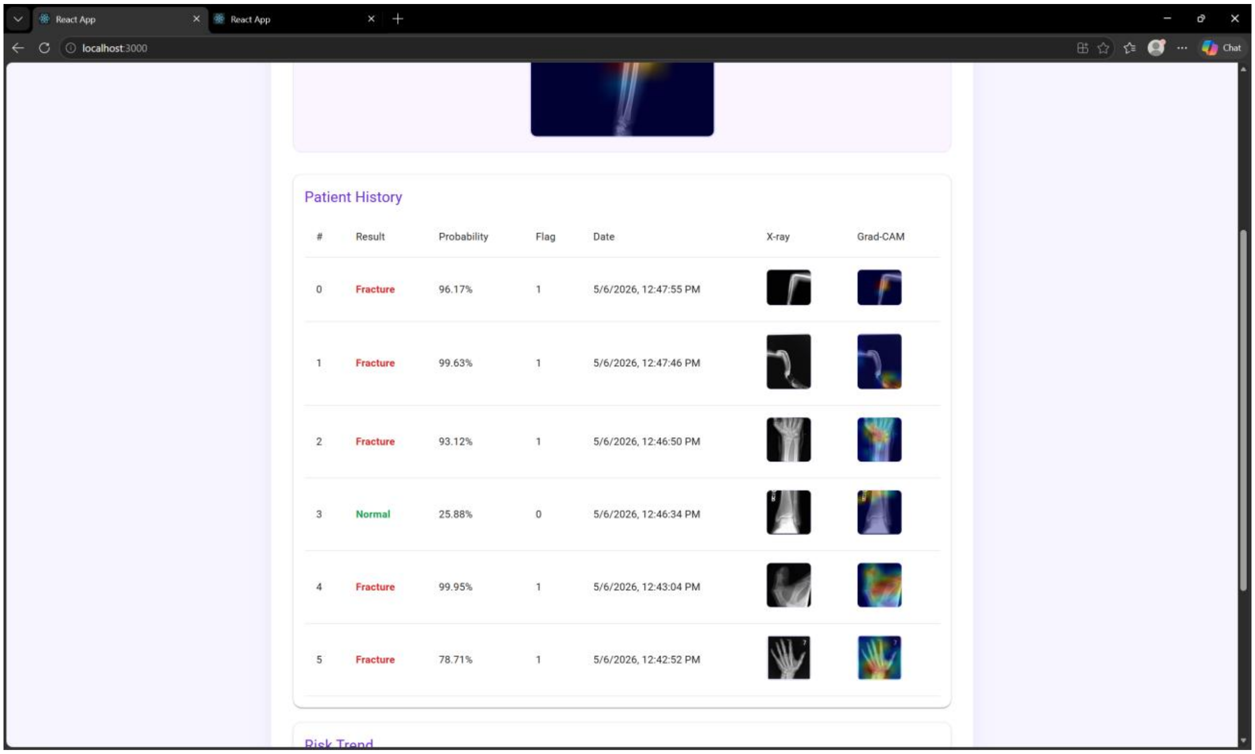
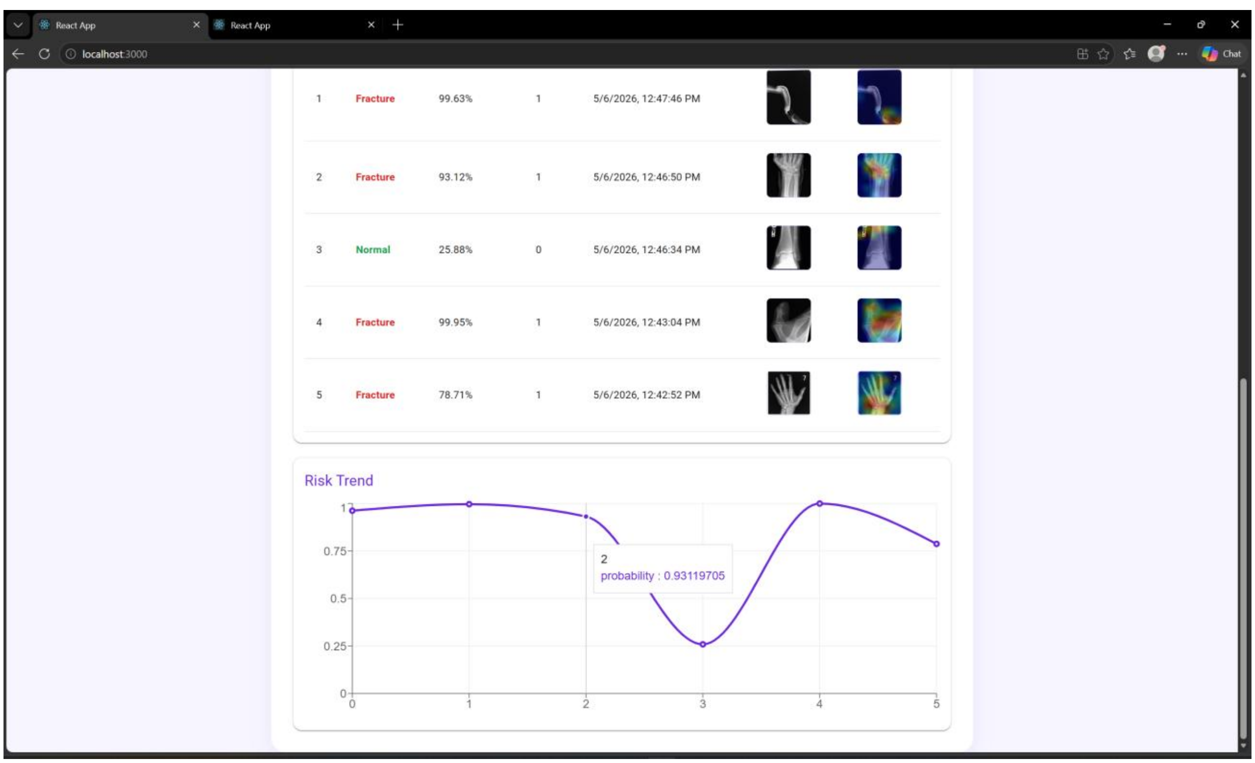

Administrator: Windows PowerShell
Epoch 9/10
93/93 - 6s - 67ms/step - accuracy: 0.9715 - loss: 0.2538
Epoch 10/10
93/93 - 6s - 68ms/step - accuracy: 0.9736 - loss: 0.2289
Evaluating on UNSEEN test set...
2/2 1s 131ms/step
=== FINAL TEST RESULTS ===
precision recall f1-score support
0.0 0.67 0.67 0.67 3
1.0 0.98 0.98 0.98 61
accuracy 0.97 64
macro avg 0.83 0.83 0.83 64
weighted avg 0.97 0.97 0.97 64
AUC: 0.9891
F1: 0.9836
Confusion Matrix:
[[ 2 1]
[ 1 60]]
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_model(model, 'my_model.keras')`.
Model saved to: C:\Users\kates\Desktop\HealthcareProject\backend\models\final_model.h5
(venv) PS C:\Users\kates\Desktop\HealthcareProject\backend>
import os
import numpy as np
import tensorflow as tf
from tensorflow.keras.applications import EfficientNetB0
from tensorflow.keras.applications.efficientnet import preprocess_input
from tensorflow.keras.layers import GlobalAveragePooling2D, BatchNormalization, Dropout, Dense, Input
IMG_SIZE = 224
BATCH_SIZE = 16
EPOCHS = 10
FOLDS = 5
fracture_class_ids = {0,1,3,4,5,6,7,8,9}
# ============== AUGMENTATION ==============
train_aug = A.Compose([
A.HorizontalFlip(p=0.5),


React App
localhost:3000
X-RAY / MRI FRACTURE DASHBOARD
Patient ID
111223
Choose File
IMG0000025.jpg
Upload & Predict
Load History
Health AI Agent
Fracture (96.17%)
Patient History
# Result Probability Flag Date X-ray Grad-CAM
0 Fracture 96.17% 1 5/6/2026, 12:47:55 PM
1 Fracture 99.63% 1 5/6/2026, 12:47:46 PM
2 Fracture 93.12% 1 5/6/2026, 12:46:50 PM


React App
React App
localhost:3000
Chat
Patient History
# Result Probability Flag Date X-ray Grad-CAM
0 Fracture 96.17% 1 5/6/2026, 12:47:55 PM
1 Fracture 99.63% 1 5/6/2026, 12:47:46 PM
2 Fracture 93.12% 1 5/6/2026, 12:46:50 PM
3 Normal 25.88% 0 5/6/2026, 12:46:34 PM
4 Fracture 99.95% 1 5/6/2026, 12:43:04 PM
5 Fracture 78.71% 1 5/6/2026, 12:42:52 PM


React App
React App
localhost:3000
Chat
1 Fracture 99.63% 1 5/6/2026, 12:47:46 PM
2 Fracture 93.12% 1 5/6/2026, 12:46:50 PM
3 Normal 25.88% 0 5/6/2026, 12:46:34 PM
4 Fracture 99.95% 1 5/6/2026, 12:43:04 PM
5 Fracture 78.71% 1 5/6/2026, 12:42:52 PM
Risk Trend
1
0.75
0.5
0.25
0
0 1 2 3 4 5
2
probability : 0.93119705

## Health AI Agent

111223

PATIENT HAD WHEN DETECTING NORMAL SCAN

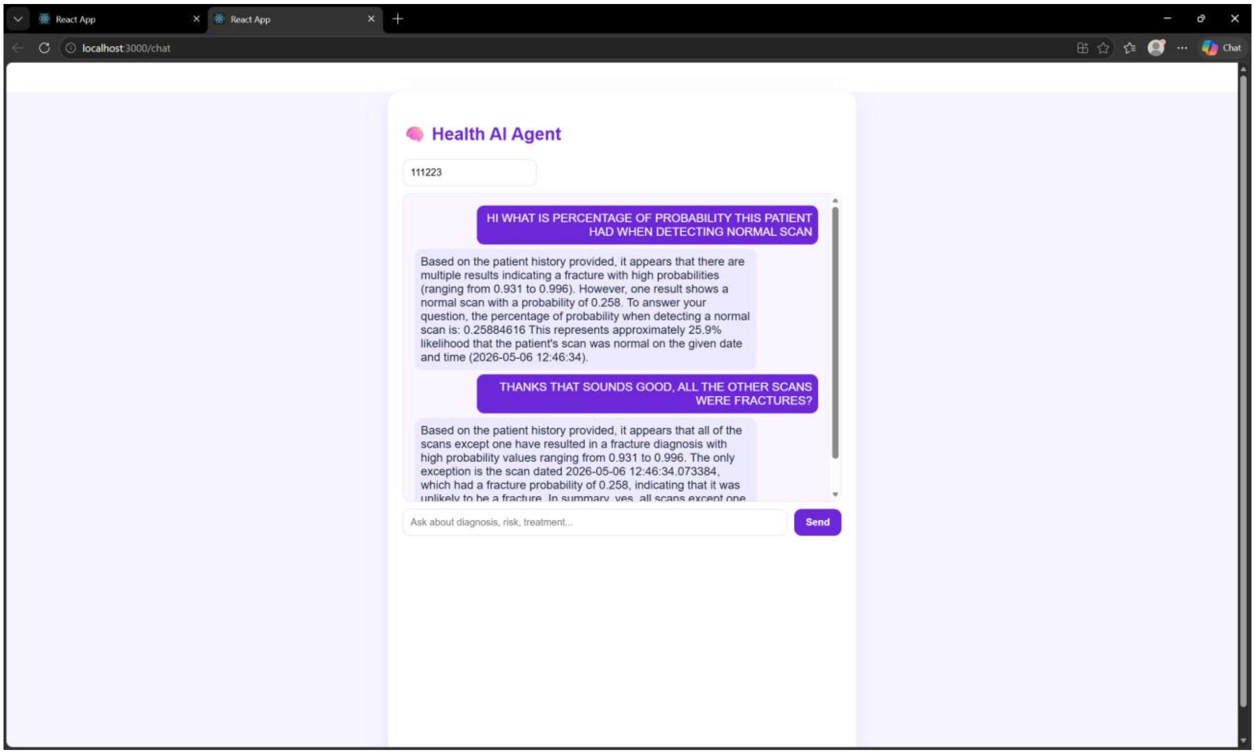
Based on the patient history provided, it appears that there are multiple results indicating a fracture with high probabilities (ranging from 0.931 to 0.996). However, one result shows a normal scan with a probability of 0.258. To answer your question, the percentage of probability when detecting a normal scan is: 0.25884616 This represents approximately 25.9% likelihood that the patient's scan was normal on the given date and time (2026-05-06 12:46:34).

THANKS THAT SOUNDS GOOD, ALL THE OTHER SCANS WERE FRACTURES?

Based on the patient history provided, it appears that all of the scans except one have resulted in a fracture diagnosis with high probability values ranging from 0.931 to 0.996. The only exception is the scan dated 2026-05-06 12:46:34.073384, which had a fracture probability of 0.258, indicating that it was unlikely to be a fracture. In summary, yes, all scans except one have been diagnosed as fractures.

Ask about diagnosis, risk, treatment...

Send


React App
React App
localhost:3000/chat
Chat
Health AI Agent
111223
HI WHAT IS PERCENTAGE OF PROBABILITY THIS PATIENT HAD WHEN DETECTING NORMAL SCAN
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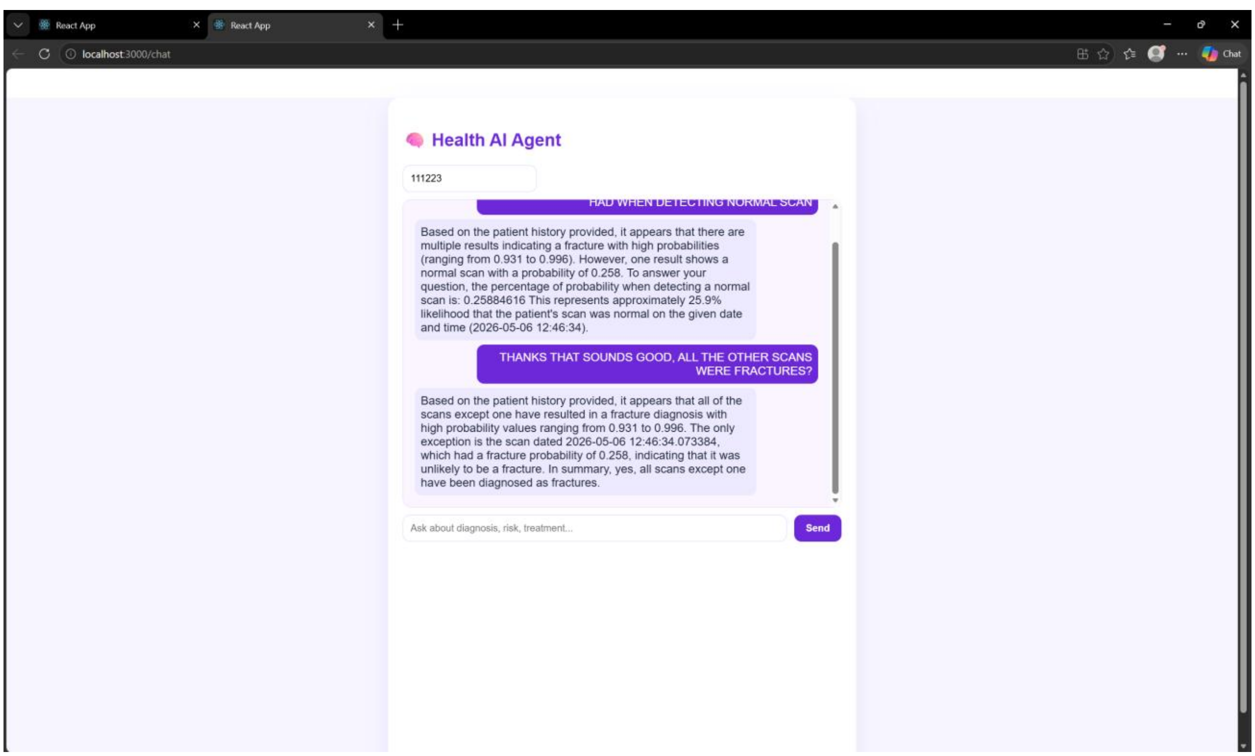
React App
React App
localhost:3000/chat
Chat
Health AI Agent
111223
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Send